22MDC56 – MACHINE LEARNING LABORATORY

SUMMARY ON RESEARCH ARTICLES

PREDICTIVE MAINTENANCE AND MACHINE FAILURE PREDICTION USING MACHINE LEARNING

Team Members:

John Silva. S [2403717673721023]

John Joeual. C [2403717673721022]

Darshan. S [2403717673721009]

LITERATURE SURVEY FOR "PREDICTIVE MAINTENANCE AND MACHINE FAILURE PREDICTION USING MACHINE LEARNING"

[1] Carvalho et al. presented a systematic literature review on machine learning methods used in predictive maintenance. The study explains different ML algorithms, challenges, and opportunities in industrial maintenance. It helped us understand the overall workflow of predictive maintenance and the importance of comparing multiple machine learning models instead of relying on a single algorithm.

[2] Stephan Matzka introduced the AI4I 2020 Predictive Maintenance dataset along with an explainable artificial intelligence approach. The dataset contains machine operating parameters such as air temperature, process temperature, rotational speed, torque, and tool wear for predicting machine failures. This paper inspired us to use the AI4I dataset as the primary dataset for our project.

[3] Surucu, Gadsden, and Yawney reviewed condition monitoring using machine learning techniques. Their research highlighted the importance of preprocessing, feature selection, and choosing suitable ML models for reliable fault detection. This article motivated us to perform detailed preprocessing and exploratory data analysis before model building.

[4] A predictive maintenance study in manufacturing systems proposed AI-based early failure detection using machine learning, deep learning, and hybrid learning models. The research demonstrated that predictive models can identify possible machine failures before production downtime occurs. It gave us the idea of developing an early warning machine failure prediction system.

[5] Fordal et al. developed a predictive maintenance platform using sensor data and artificial neural networks for Industry 4.0 applications. The paper showed how sensor information can be transformed into meaningful maintenance predictions for industrial equipment. This inspired us to design a web-based predictive maintenance interface using machine operating parameters.

[6] Serradilla et al. reviewed deep learning models for predictive maintenance and compared different architectures used for anomaly detection, fault diagnosis, and remaining useful life prediction. The study helped us understand the advantages and limitations of deep learning compared with traditional machine learning approaches and encouraged us to begin with interpretable ML models before extending to advanced techniques.

[7] The A^2-LSTM study proposed an attention-based LSTM model for predictive maintenance of industrial equipment. It emphasized learning important sensor attributes and reducing redundant information while

predicting equipment conditions. This research gave us knowledge about advanced feature learning techniques that can be considered as future enhancements to our project.

[8] A real-world automotive predictive maintenance study focused on data-driven strategies, feature engineering, signal selection, and validation methods for machine health prediction. The paper emphasized that meaningful feature engineering significantly improves predictive performance. This motivated us to analyze relationships between temperature, torque, rotational speed, and tool wear during EDA.

[9] Guidotti, Pandolfo, and Pulina conducted a systematic review of supervised machine learning techniques for predictive maintenance in Industry 4.0. The study analyzed numerous industrial applications and highlighted the importance of open datasets, explainable AI, and reproducible machine learning experiments. This article influenced our decision to compare several supervised classification algorithms using the AI4I dataset.

[10] Recent research on explainable AI for industrial fault diagnosis emphasized that high-performing machine learning models should also provide understandable explanations for maintenance decisions. The study discussed the importance of transparency and trust when applying AI in industrial environments. This inspired us to include feature importance and explainable prediction as business insights in our proposed system.

GAP ANALYSIS:

Most existing research works focus on improving prediction performance using individual machine learning, deep learning, or hybrid models. Some studies emphasize anomaly detection, while others concentrate on explainable AI or industrial sensor analysis. However, many predictive maintenance implementations mainly evaluate model accuracy without combining preprocessing, exploratory data analysis, model comparison, explainability, and a user-friendly deployment in one complete system.

The proposed project aims to address this gap by developing an end-to-end Predictive Maintenance System using the AI4I 2020 Predictive Maintenance Dataset. The system will perform data preprocessing, exploratory data analysis, feature engineering, comparison of multiple machine learning classification models, machine failure prediction, feature importance analysis, and deployment through a simple web interface. The objective is to provide both accurate predictions and meaningful maintenance insights for industrial applications.

REFERENCES:

1. T. P. Carvalho, F. A. A. M. N. Soares, R. Vita, R. P. Francisco, J. P. T. V. Basto and S. G. S. Alcalá, "A Systematic Literature Review of Machine Learning Methods Applied to Predictive Maintenance," Computers & Industrial Engineering, vol. 137, 2019. DOI: 10.1016/j.cie.2019.106024.

2. S. Matzka, "Explainable Artificial Intelligence for Predictive Maintenance Applications," 2020 Third International Conference on Artificial Intelligence for Industries (AI4I), pp. 69–74, 2020. DOI: 10.1109/AI4I49448.2020.00023.

3. O. Surucu, S. A. Gadsden and J. Yawney, "Condition Monitoring using Machine Learning: A Review of Theory, Applications, and Recent Advances," Expert Systems with Applications, vol. 221, 2023. DOI: 10.1016/j.eswa.2023.119738.

4. "A Predictive Maintenance Approach in Manufacturing Systems via AI-Based Early Failure Detection," Manufacturing Letters, vol. 35, pp. 1179–1186, 2023. DOI: 10.1016/j.mfglet.2023.08.125.

5. J. M. Fordal, P. Schjølberg, H. Helgetun, T. Ø. Skjermo, Y. Wang and C. Wang, "Application of Sensor Data Based Predictive Maintenance and Artificial Neural Networks to Enable Industry 4.0," Advances in Manufacturing, vol. 11, pp. 248–263, 2023.

6. O. Serradilla, E. Zugasti and U. Zurutuza, “Deep Learning Models for Predictive Maintenance: A Survey, Comparison, Challenges and Prospect,” 2020.

7. “A²-LSTM for Predictive Maintenance of Industrial Equipment Based on Machine Learning,” Computers & Industrial Engineering, vol. 172, 2022. DOI: 10.1016/j.cie.2022.108560.

8. “Data-Driven Strategies for Predictive Maintenance: Lessons Learned from an Automotive Use Case,” Computers in Industry, vol. 134, 2022. DOI: 10.1016/j.compind.2021.103554.

9. D. Guidotti, L. Pandolfo and L. Pulina, “A Systematic Literature Review of Supervised Machine Learning Techniques for Predictive Maintenance in Industry 4.0,” IEEE Access, vol. 13, pp. 102479–102504, 2025. DOI: 10.1109/ACCESS.2025.3578686.

10. J. Cação, J. Santos and M. Antunes, “Explainable AI for Industrial Fault Diagnosis: A Systematic Review,” Journal of Industrial Information Integration, vol. 47, 2025. DOI: 10.1016/j.jii.2025.100905.